PSEUDO-COHERENCE OF A MORPHISM IS FPPF LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 17626-17677, immediately following the statement at 17620-17624.

Editorial note. The author states the remaining algebra problem explicitly ('the algebra problem is the following: ...') inside the span; it is retained verbatim.

Editorial note. Three cross-chapter pointers (Descent, More on Algebra) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Dominating an fppf covering by a nicer one. chapter lemma lemma-dominate-fppf, cited at proof line 17630. Replaces the given fppf covering by one with a Zariski covering of the source and affine pieces, which makes the algebra question accessible.
- Composition and flat finite presentation inputs. chapter lemmas lemma-composition-pseudo-coherent and lemma-flat-finite-presentation-pseudo-coherent, cited in the proof. Controls pseudo-coherence under composition and under flat morphisms of finite presentation.
- Permanence of pseudo-coherence. chapter lemma lemma-permanence-pseudo-coherent with Descent, lemma descent-lemma-flat-finitely-presented-permanence, cited in the proof. Moves pseudo-coherence along the flat finitely presented pieces of the covering.
- Algebraic ingredients on pseudo-coherence of ring maps. More on Algebra, lemmas more-algebra-lemma-finite-extension-pseudo-coherent and more-algebra-lemma-summands-relative-pseudo-coherent, cited at proof lines 17669-17675. Solves the resulting ring-theoretic problem: if R -> A' is pseudo-coherent and A' is a free A-module of finite rank, then R -> A is pseudo-coherent.

Source reference. Original reference 02KL: focused-seeds/geometry/S9-morph-pseudo-coherent-fppf-source/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0681: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17402–17402. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05WN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13649–13649. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0695: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17428–17428. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0683: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17440–17440. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0673: focused-seeds/geometry/S9-morph-pseudo-coherent-fppf-source/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0676: focused-seeds/geometry/S9-morph-pseudo-coherent-fppf-source/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *The property $\mathcal{P}(f) =$ "f is pseudo-coherent" is fppf local on the source.*

Proof. Let $f : X \to S$ be a morphism of schemes. Let $\{g_i : X_i \to X\}$ be an fppf covering such that each composition $f \circ g_i$ is pseudo-coherent. According to Lemma 05WN there exist

(1) a Zariski open covering $X = \bigcup U_j$,
(2) surjective finite locally free morphisms $W_j \to U_j$,
(3) Zariski open coverings $W_j = \bigcup_k W_{j,k}$,
(4) surjective finite locally free morphisms $T_{j,k} \to W_{j,k}$

such that the fppf covering $\{h_{j,k} : T_{j,k} \to X\}$ refines the given covering $\{X_i \to X\}$. Denote $\psi_{j,k} : T_{j,k} \to X_{\alpha(j,k)}$ the morphisms that witness the fact that $\{T_{j,k} \to X\}$ refines the given covering $\{X_i \to X\}$. Note that $T_{j,k} \to X$ is a flat, locally finitely presented morphism, so both X_i and $T_{j,k}$ are pseudo-coherent over X by Lemma 0695. Hence $\psi_{j,k} : T_{j,k} \to X_i$ is pseudo-coherent, see Lemma 0683. Hence $T_{j,k} \to S$ is pseudo coherent as the composition of $\psi_{j,k}$ and $f \circ g_{\alpha(j,k)}$, see Lemma 0681. Thus we see we have reduced the lemma to the case of a Zariski open covering (which is OK) and the case of a covering given by a single surjective finite locally free morphism which we deal with in the following paragraph.

Assume that $X' \to X \to S$ is a sequence of morphisms of schemes with $X' \to X$ surjective finite locally free and $X' \to Y$ pseudo-coherent. Our goal is to show that $X \to S$ is pseudo-coherent. Note that by Descent, Lemma 02KL the morphism $X \to S$ is locally of finite presentation. It is clear that the problem reduces to the case that X', X and S are affine and $X' \to X$ is free of some rank $r > 0$. The corresponding algebra problem is the following: Suppose $R \to A \to A'$ are ring maps such that $R \to A'$ is pseudo-coherent, $R \to A$ is of finite presentation, and $A' \cong A^{\oplus r}$ as an A-module. Goal: Show $R \to A$ is pseudo-coherent. The assumption that $R \to A'$ is pseudo-coherent means that A' as an A'-module is pseudo-coherent relative to R. By More on Algebra, Lemma 0673 this implies that A' as an A-module is pseudo-coherent relative to R. Since $A' \cong A^{\oplus r}$ as an A-module we see that A as an A-module is pseudo-coherent relative to R, see More

on Algebra, Lemma 0676. This by definition means that $R \to A$ is pseudo-coherent and we win. $\square$